

The role of entropy of review text sentiments on online WOM and movie box office sales



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ABSTRACT

Sentiments from online word-of-mouth (WOM) are often controversial, since individuals have different preferences toward the same products. Past studies have focused on online WOM effects by measuring WOM volume and valence. However, few studies have investigated how the entropy of the review text sentiment influences the relationship between online WOM and product sales. As WOM valence and volume are usually provided at an aggregated level, consumers often do not have enough information to make a decision. In this case, reading online review text has become an important process for consumers to make purchasing decisions. However, when consumers encounter too many positive review texts, they may doubt the credibility of online WOM itself. Thus, we analyzed the entropy of the review text sentiments by conducting text-mining techniques. We classified review text sentiment into positive, negative, and neutral categories and created an entropy variable. A high level of entropy in review texts indicates that sentiment from review texts are equally distributed, but not biased, towards positive or negative sentiment. We estimated our research model with the entropy variable in a panel dataset for 204 movies over a half-year period. The results suggest that the entropy level in the review texts has a positive moderating impact on the relationship between WOM (e.g., valence and volume) and movie box office sales. The findings imply that deleting negative reviews to enhance product sales may not help online retailers or related parties.

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1. Introduction

Word-of-mouth (WOM) has been regarded as one of the most effective information sources, particularly for experience goods (Godes and Mayzlin, 2004; Granovetter, 1973). As new information technology enables people to exchange online opinions more freely and easily without the limitations of time and space, it has become important for firms to manage venues and the spread of WOM (e.g., online user review systems), to attract more consumers. As consumers have begun to share their opinions through the Internet, online WOM has been found to influence millions of web surfers in their decisions. That is, many potential consumers typically seek out online reviews or message boards to learn about the experiences of others to make better purchasing decisions.

In particular, online WOM has become popular in the motion picture industry. Since movies are an experience good, potential consumers may have difficulty in evaluating product quality. In this regard, consumers usually search for past consumer opinions. Past studies have examined the relationship between online WOM

and movie sales (Basuroy et al., 2003; Duan et al., 2008a,b; Liu, 2006), and showed that WOM remains a powerful source for movie consumers. Additionally, we often observe that some movies produced with much smaller budgets have achieved great success by utilizing viral marketing through online WOM (e.g., *Cloverfield*, *The Dark Night*).

Early studies in the literature have investigated the impact of WOM valence (e.g., average movie ratings) and volume (e.g., total number of reviews) on product sales. These studies have mainly used WOM valence and volume at an aggregated level. However, many researchers have recognized that such an aggregated WOM may not provide consumers with enough information to make a product purchase choice (Bai, 2011; Cao et al., 2011; Chen et al., 2008; Chen and Lurie, 2013; Ghiassi et al., 2013; Ghose and Ipeirotis, 2011; Li and Wu, 2010; Li et al., 2013; Moore, 2015; Packard and Berger, 2016; Yin et al., 2014, 2017). For example, consumers may need to choose between two movies that have almost identical user ratings and a number of reviews (e.g., In the Naver Movie, the user rating value of “Avengers: Age of Ultron” was 8.56, while that of “Mad Max: Fury Road” was 8.83). In this situation, consumers need more information to make a choice between the two movies. That is, they may want to examine the review

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texts and read them carefully. However, consumers may doubt the credibility of user rating values when they encounter too many positive reviews. More specifically, they may speculate that firms are hiring people to delete negative reviews (or to post positive reviews).

Also, it is natural for consumers to expect various types of review text sentiments since a movie is an experience good. Although user ratings couldn't reflect the various types of sentiment as an aggregated level of information, consumers may perceive more credibility in user ratings when they encounter an equally distributed review text sentiment. Since users may have different preferences towards the same movie, an equally distributed review text sentiment makes sense for an experience good (e.g., a movie). Therefore, when positive, negative, and neutral types of review text sentiment are equally distributed in online review pools, the mixed review text sentiment may give more credibility to the online WOM variables (e.g., user ratings and number of reviews). Such a phenomenon can be explained by the dilution effect.

The dilution effect refers to irrelevant or neutral information's weakening effect on individual stereotypes (Nisbett et al., 1981). Stereotypes may lead people to illustrate a biased judgment. However, these stereotypes can be reduced by providing people with irrelevant, or other types, of sentiment about stereotyped people or goods. In our research context, consumers may have a stereotype towards online WOM information and doubt the credibility of movie reviews, as many related parties are known to censor or manipulate movie reviews. However, if negative, neutral, irrelevant, or other types of review text sentiment are added to positive review pools, such a stereotype of review sentiment can be weakened. That is, the impact of online WOM on movie sales can be enhanced when valence and volume are interacted with an evenly distributed set of review text sentiment.

We classified review texts into positive, negative, and neutral categories and created an entropy variable to measure the level of heterogeneity in review texts. A high level of entropy in review texts indicated that the sentiment in the reviews were equally distributed, but were not biased towards either positive or negative reviews. We collected WOM and sales data for 204 movies over a half-year period (from July 2012 to Jun 2013), and established a panel to estimate our research model. In our research model, we included the entropy of the review text sentiment as a key variable and then conducted a panel estimation. The results were consistent with our expectation that entropy level is positively moderated with WOM valence and volume, which influences movie box office sales.

Our findings have several implications. First, our study contributes to the related literature in that our results show that the entropy of the review text sentiment positively moderates with the WOM valence and volume when influencing product sales. That is, when the sentiment in the reviews are well distributed, the impact of WOM valence (or volume) on movie sales becomes greater. Because the existing studies have not investigated a moderating effect between review text sentiments and WOM information, we believe that this is the first study to investigate how the entropy of the review text sentiment is moderated with WOM values to influence product sales. Second, we note that some previous studies empirically illustrated that WOM valence (e.g., rating values) is positively associated with product sales (Clemons et al., 2006; Dellarocas et al., 2007), while others showed that WOM valence does not have any impact on product sales (Duan et al., 2008; Liu, 2006; Gu et al., 2012). Due to these mixed results, many researchers have recognized the limitation of the aggregated numerical review scores (e.g., WOM valence). Thus, the literature has tried to disaggregate and find what type of online reviews can be evaluated as useful or helpful by potential consumers

(Chen et al., 2008; Forman et al., 2008; Mudambi and Schuff, 2010; Cao et al., 2011; Schindler and Bickart, 2012; Chen and Lurie, 2013; Li et al., 2013; Moore, 2015; Yin et al., 2014, 2017). Unlike these studies, however, we believe that the mixed results may come from the lack of consideration of the review text sentiments and their moderating relationship with WOM information.

Our results also contribute to the related literature regarding practical application. It is frequently reported that many online retailers try to eliminate negative reviews by censoring them or by including more positive reviews. For example, related parties may want to increase average rating values, and thus, enhance their movie sales by deleting negative reviews. However, our results illustrate that average rating values statistically do not have any impact on product sales. In addition, we show that the WOM valence become influential when it interacts with a high level of entropy value. Thus, our results indicate that censoring negative ratings just decreases the entropy value, which leads to a less influential WOM effect. Therefore, from a managerial perspective, deleting negative reviews or including more positive reviews may not help online retailers or related parties.

The rest of the paper is organized as follows. The next chapter presents the literature review; this is followed by the research hypotheses. Subsequently, we provide an explanation for our data source and the empirical model. Finally, the conclusions are presented. The paper ends with a discussion of the limitations and future research ideas.

2. Literature review

2.1. Online WOM effects

Past studies demonstrate that online WOM has a significant influence on the retail sales of various consumer goods (e.g., Chatterjee, 2001; Clemons et al., 2006; Godes and Mayzlin, 2004; Liu, 2006; Zhu and Zhang, 2006; King et al., 2014). These online WOM studies have been extended into the movie industry, where many researchers have investigated the relationship between online WOM and movie box office sales, as they could easily get movie box office sales data (Dellarocas et al., 2007; Duan et al., 2008a,b). In particular, Elberse and Eliashberg (2003) used revenue per screen during the previous week as a proxy of WOM in their analysis of the supply and demand of movies, suggesting that such a measurement of WOM is a key predictor of box office revenue. Dellarocas et al. (2007) suggested a modified Bass Diffusion model to investigate the impact of online user reviews in forecasting movie revenues. These studies have demonstrated the positive relationship between movie sales and WOM information. They have applied various empirical models to examine WOM effects. As a result, firms and organizations consider online WOM an important source of business strategies (Brynjolfsson et al., 2003; Chen and Xie, 2008; Dellarocas, 2003).

The previous studies in the literature have used customer rating values as an indicator of WOM valence. However, other studies insist that such user ratings do not have a significant direct effect on movie sales (Liu, 2006; Duan et al., 2008a). These studies focus on the daily variation of product sales, as compared to earlier research that investigated weekly variation (Neelamegham and Chintagunta, 1999; Elberse and Eliashberg, 2003; Liu, 2006). Eliashberg and Shugan (1997) illustrated that movie critic ratings are predictors of movie performance, but do not influence movie performance. Duan et al. (2008a) employed user rating values to observe the WOM valence effect and tried to capture the endogenous nature between movie box office sales and WOM volume. They argued that WOM is not only a driving force behind consumer purchases, it is also an outcome of product sales, where a causality

exists between product sales and WOM. More specifically, [Duan et al. \(2008a\)](#) used a three-stage least squares (3SLS) estimation to control for endogeneity problems by constructing a simultaneous equation system. However, they found that the coefficient of WOM valence was not statistically significant. [Gu et al. \(2012\)](#) also showed that user ratings from a retailer-hosted online WOM do not have a significant impact on product sales. They tried to differentiate the impact of an external WOM and that of an internal WOM for high-involvement products. They hypothesized that an external WOM had a greater influence on the sales of high-involvement products at online retailers than a retailer-hosted internal WOM. They empirically illustrated the dominant influence of the external WOM over the internal WOM. In the line of these studies, we try to provide one of the possible answers to explain why online WOM valence (e.g., user rating values) does not have a significant impact on product sales by investigating the moderating effects of the entropy of the review text sentiment.

Due to these mixed results of online WOM effects, many researchers have recognized the limitations of aggregated numerical review scores (e.g., WOM valence). That is, the literature has already tried to disaggregate and find what types of online reviews can be evaluated as useful or helpful by potential consumers. Thus, many researchers have paid attention to helpful reviews that may have a stronger impact on consumers' choices. For instance, [Chen et al. \(2008\)](#) empirically showed that the proportion of helpful votes on product reviews can serve as an indicator of the quality of reviews to potential consumers, and some prior studies have examined what factors influence review helpfulness by examining review characteristics, reviewer identity, product types, the number of votes etc. ([Cao et al., 2011](#); [Chen et al., 2008](#); [Forman et al., 2008](#); [Mudambi and Schuff, 2010](#)). The previous research also noted that negative reviews usually have a greater impact on consumers' choices than positive reviews ([Basuroy et al., 2003](#); [Charlett et al., 1995](#); [Chevalier and Mayzlin, 2006](#); [Herr et al., 1991](#); [Mizerski, 1982](#)). In line with these studies, many researchers have examined the linguistic content of reviews through lab and field experiments ([Chen and Lurie, 2013](#); [Li et al., 2013](#); [Moore, 2015](#); [Schindler and Bickart, 2012](#); [Packard and Berger, 2016](#); [Yin et al., 2014, 2017](#)). [Li et al. \(2013\)](#) tried to define helpful reviews and suggested three dimensions of review helpfulness – perceived source credibility, perceived content diagnosticity, and perceived vicarious expression – and showed that consumers perceive a concrete review as more helpful than an abstract review ([Schellekens et al., 2010](#)). [Schindler and Bickart \(2012\)](#) investigated the impact of content and message style on perceived helpfulness. In particular, [Yin et al. \(2014\)](#) examined the relationship between the emotion of a review and its perceived helpfulness and further showed the non-linear relationship between emotional arousal and helpfulness ([Yin et al., 2017](#)). Additionally, [Moore \(2015\)](#) showed this relationship via the explanation type of review (e.g., actions vs. reactions). Recently, [Yin et al. \(2016\)](#) showed how confirmation bias influences perceived helpfulness depending on consumers' prior beliefs regarding average rating values.

Another stream of this research that goes beyond WOM valence and volume has utilized text-mining techniques. Many researchers have examined online sentiments by using these techniques ([Bai, 2011](#); [Cao et al., 2011](#); [Ghiassi et al., 2013](#); [Ghose and Ipeirotis, 2011](#); [Li and Wu, 2010](#)). For example, [Li and Wu \(2010\)](#) examined the identification of online forum hotspots, where helpful opinions were exposed to other members by using sentiment analysis based on K-means clustering and a support vector machine (SVM). [Cao et al. \(2011\)](#) investigated the “helpfulness” feature of online user reviews by using the Latent Semantic analysis (LSA)-based text-mining methodology. In addition, [Ghose and Ipeirotis \(2011\)](#) proposed techniques for predicting the helpfulness and importance

of a review by integrating econometric, text-mining, and predictive modeling techniques.

Our study is closely related to these studies that go beyond WOM valence and volume, but ours is different in that we examine the relationship between the entropy of the review text sentiment and online WOM effects in the movie industry. In particular, we provide the first large-sample evidence on this issue with 204 movies and 256,998 reviews from the largest online portal service provider in South Korea. Unlike previous studies, this study uses the text-mining technique to classify review texts and sees how review texts are related to WOM valence and volume. While prior research has recognized the importance of the linguistic content of reviews, a consideration of text-mining techniques to examine the content of reviews carefully has been lacking in the literature. Thus, we believe that our study makes a contribution to the related literature as it is the first to show how the entropy of the review text sentiment is positively moderated by WOM information (e.g., valence or volume) to influence movie box office sales.

2.2. Text sentiments and entropy

In online user review sites, consumers share their experiences and satisfaction with experience goods (e.g., movies). This has been shown to foster the spread of information. Online reviews provide useful and relevant information for a consumer in the movie market, as they cannot evaluate movie quality prior to actual consumption. Although online reviews are generally viewed as a source of vital information in the consumption of movies, people often claim to have obtained information not relevant to movies, including thru celebrity and film gossip. There is a great deal of evidence indicating that the presence of irrelevant information can affect product evaluation and consumer choice adversely in a variety of situations ([Simonson et al., 1993](#); [Van Osselaer et al., 2003](#)). But while it is clear that people communicate frequently with others through online reviews, and that such online communication plays a significant role in shaping consumer behavior, less is known about how neutral information might affect consumer purchasing decisions, particularly in many experience good markets.

In this study, our model assesses the specific role of online reviews on movie box-office performance after accounting for various types of review text sentiment (e.g., positive, negative, or neutral ones). To categorize review text sentiment, we used a text-mining technique, which was applied to extract three clusters of positive, negative, and neutral review text sentiment. We borrow the concept of entropy, as first introduced by [Shannon \(1948\)](#), to measure the relevant (e.g., positive and negative reviews) and other (neither positive nor negative) review text effects. In “*A Mathematical Theory of Communication*,” [Shannon \(1948\)](#) argued that entropy represents uncertainty, which can reach a maximum value when each independent event displays the same probability. Additionally, [Park et al. \(2012\)](#) incorporated the concept of entropy into their empirical model, suggesting that entropy is frequently implemented as a proxy to represent the level of complexity. Other recent studies utilized the entropy measure to select top-k user profiles with a higher degree of certainty ([Abbasi et al., 2008](#); [Ayanso et al., 2009](#); [Bruno et al., 2002](#)). In this study, entropy was also used in a similar manner to measure the ratio of the review text sentiment.

[Fig. 1](#) illustrates the relationship between entropy and probability in the case of a coin toss. As we see in [Fig. 1](#), entropy has a maximum value when heads and tails both have an equal probability of 1/2, or fair odds. This implies that every single event with fair conditions maximizes uncertainty, because we cannot easily predict which side of the coin will appear on the next toss. In contrast, if a coin is double-headed or double-tailed and only comes up on just one side, there is no uncertainty, and entropy becomes zero. That

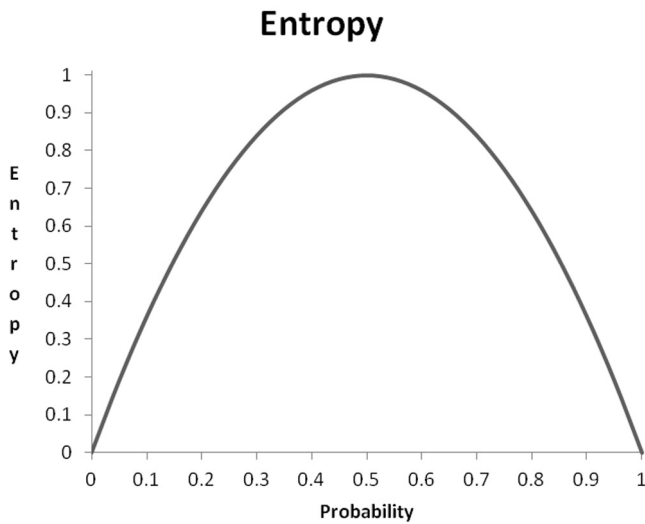


Fig. 1. The shape of information entropy.

is, the more entropy grows, the more uncertainty arises. Greater uncertainty may indicate a plethora of information, which is appropriate in the selection of experience goods.

We adopted this entropy theory for our research to measure the ratio of the review text sentiment. We found that, in addition to positive and negative reviews, there are also neutral reviews that do not indicate any particular direction. Thus, we calculated the entropy for three segments: positive, negative, and neutral reviews. Our primary goal was to investigate how the distribution of the review text sentiment influences the relationship between online WOM and movie box office sales. As consumers encounter positive, negative, and neutral types of review texts, they may come to have an increased trust in the value of online WOM (e.g., user rating values). Previous studies also suggested that negative reviews may enhance the credibility of online reviews. [Awad and Zhang \(2007\)](#) demonstrated that WOM has a greater influence on product sales when the retailer does not censor negative consumer reviews. [Sorensen and Rasmussen \(2004\)](#) examined the impact of New York Times book reviews on sales. Their results suggest that even negative reviews lead to increases in sales. We complement these studies by measuring the entropy of the review text sentiment effects. Thus, this study contributes to the related literature in that the entropy of the review text sentiment may increase online WOM effects on product sales.

3. Research hypotheses

The main purpose of this study is to investigate the impact of the entropy of the review text sentiment on the relationship between online WOM and movie box office sales. We note that some previous studies empirically showed that WOM valence (e.g., rating values) is positively associated with product sales ([Clemons et al., 2006](#); [Dellarocas et al., 2007](#)), while the others illustrated that WOM valence does not have an impact on product sales ([Duan et al., 2008](#); [Liu, 2006](#); [Gu et al., 2012](#)). We argue that the mixed results may come from the lack of consideration of the review text sentiment and their moderating relationship with WOM information. Thus, this is the first study to investigate how the entropy of the review text sentiment moderates with WOM information to influence product sales.

Consumers at times may find it difficult to make purchase decisions based only on WOM volume and WOM valence when the decision may be between two movies that have almost identical

user rating values and the total number of reviews. For example, assume that one movie consumer is a big fan of a science fiction (SF) movie and wants to watch any SF movie this Friday night. To decide which SF movie to watch, she or he may first search for WOM information. That is, she or he may take a look at the user rating values and the total number of reviews to make a movie choice. However, sometimes consumers may notice that user rating values and the number of reviews are almost identical across several SF movies¹. Therefore, they may want to obtain additional information to make a decision. In this case, consumers may try to read the review texts carefully for each movie. That being said, it is natural for consumers to expect various types of review text sentiment, since a movie is an experience good.

This example suggests that numeric user ratings or the number of reviews for a movie couldn't reflect the various types of sentiment as an aggregated level of information, while review texts can produce various types of sentiment. This may suggest that the distribution of rating values is different with that of review text sentiment. To explain this inconsistency, we note the theory of information cascade. Information cascades refer to the phenomenon of following the other's result of activity, who was in a similar decision making condition, instead of considering one's own information in the decision making process. That is, information cascades is an action of taking on another's opinion over their own thoughts. Many previous studies confirmed that information cascades affect the choice of product (e.g., [Walden and Browne, 2002](#); [Kauffman and Li, 2003](#); [Golder and Tellis, 2004](#); [Duan et al., 2009](#)). In our research context, movie consumers might be anchored by the WOM valence, as the average rating value is saliently exposed on the top of the movie review websites. However, when they leave review texts about a movie, there is little motive to follow another opinion. Therefore, users more freely leave their opinions as a text form and show the various types of text sentiment. This suggests that the distribution of the rating values is different from those of the text-mining results.

Given these arguments above, it's natural that consumers may feel more comfortable observing an evenly distributed review text sentiment, as people have various preferences about the same movie. That is, consumers may doubt the credibility of aggregated WOM information (e.g., user ratings and the number of reviews) when encountering too many positive review texts, while they may increase the credibility of WOM information when observing evenly distributed review text sentiment. Such phenomena could be explained by the dilution effect ([Nisbett et al., 1981](#)). The dilution effect refers to irrelevant information's weakening effect on individual stereotypes ([Nisbett et al., 1981](#)). The stereotypes can be reduced by providing people with irrelevant information about the stereotyped people or goods. In our research context, users who search movie reviews may have stereotypes when they encounter too many positive reviews. Since a movie is a preference good, consumers may naturally expect evenly distributed positive, negative, or other types of reviews (e.g., neutral) about a movie. That is, they may doubt the credibility of movie reviews, as many related parties are known to censor or manipulate movie reviews. Such a stereotype can be reduced when users encounter evenly distributed positive, negative, or other types of review texts (e.g., neutral or irrelevant). Therefore, consumers may feel more comfortable and secure about WOM information when they observe evenly distributed review text sentiment, which may positively influence product sales.

¹ The average rating values for "The Avengers: Age of Ultron" and "Mad Max: Fury Road" were 8.56 and 8.83, respectively, at Naver.com. The 0.27 points difference between the two movies may not tell us which movie's quality is better. As such, users cannot decide which movie they are going to watch based on WOM valence alone. The number of reviews between the two movies were also similar.

Consequently, we expect that when consumers observe a WOM valence value of “9.0” with evenly distributed positive, negative, or other types of reviews (e.g., neutral) and a WOM valence value of “9.0” with almost all positive reviews, they are more influenced by a WOM valence with evenly mixed review text sentiment. At the same time, when consumers observe a WOM volume value of “500” with evenly distributed positive, negative, or other types of reviews (e.g., neutral) and a WOM volume value of “500” with almost all positive reviews, they are more influenced by a WOM volume with an evenly mixed review text sentiment.

In summary, movie consumers may have stereotypes towards WOM information, in particular, when they observe too many positive reviews. Such a stereotype can be weakened when consumers observe evenly distributed review text sentiment according to the dilution effect. Thus, they are more likely trust the WOM information (e.g., valence or volume) with an evenly distributed review text sentiment. That is, when WOM valence and volume are moderated with a high level of entropy in the review text sentiment, the influence of the WOM information on product sales becomes stronger.

Hence, we propose the following hypotheses²:

H1a. The entropy in the review text sentiment positively moderates the impact of the WOM valence on movie box office sales.

H1b. The entropy in the review text sentiment positively moderates the impact of the WOM volume on movie box office sales.

To measure the moderating effect of the entropy of the review text sentiment, we created interaction terms between each WOM variable and the entropy of the review text sentiment. However, we note that if consumers encounter too high a level of entropy in the review text sentiment, they may become confused. For example, consumers may place a greater weight on an online WOM when they encounter some negative and other types of review texts. However, if they observe too many negative or neutral review texts, given the same amount of positive reviews, this may decrease the credibility of the reviews. If this is the case, there is a non-linear relationship between the WOM variables and the entropy of the review text sentiment. We use the squared terms between the entropy and the WOM variables to examine their decreasing effects on movie box office sales.

Therefore, we expect that the entropy level of the review text sentiment may have a positive effect on the relationship between WOM information and movie sales, but there could be decreasing non-linear impacts. Thus, we propose the following hypotheses:

H2a. The moderating effect of the entropy of the review text sentiment on the relationship between the WOM valence and movie box office sales will decrease as the entropy value increases.

H2b. The moderating effect of the entropy of the review text sentiment on the relationship between the WOM volume and movie box office sales will decrease as the entropy value increases.

4. Data

We chose movies as our research context for several reasons. First, the movie industry has a vast number of consumers, most of which usually exchange their opinions through online platforms. Second, official box office data is easily accessible to the public and is updated daily. As a result, we can obtain reliable daily data. This may help us to better understand how WOM works in the early stages of a movie release.

The data for this study was collected from two sources: the Korean Box Office Information System (<http://www.kobis.or.kr/>) and

Naver Movie (<http://movie.naver.com/>). KOBIS, operated by the Korean Film Council (KOFIC), provides a massive amount of data related to movie box office information; this data is updated daily. Thus, we collected data on movie box office sales, the size of the audience, the number of screens, and other related control variables from KOBIS. Naver is the most popular online portal service provider in South Korea. It provides an enormous amount of information for news, sports, securities, maps, and merchandise. In October of 2014, Naver accounted for a 75% market share in online searches in South Korea. The fact that almost 75% of web surfers use Naver to obtain information enables us to make the assumption that choosing Naver Movie may generalize our results. We collected a total of 256,998 review texts from 204 movies³ at Naver Movie that were previously chosen from the KOBIS data.

To calculate our main variables, we first conducted a text-mining analysis to classify all the review texts into three categories: positive, negative, and neutral texts, with respect to the sentiment expressed in the reviews. To classify the review texts, we conducted a Latent Semantic Analysis (LSA). An LSA is a statistical method used to analyze “relationships between a set of documents and terms in these documents that produces a set of meaningful patterns related to the documents and terms (Cao et al., 2011)”. There are five steps to process review texts: text preprocessing, parsing, term reduction, singular value decomposition, and text cluster analysis. In the text preprocessing step, we simply calculated the number of words, the number of characters, their variances, and the summary statistics of the review texts. In the parsing step, we conducted text stemming by using the SAS Enterprise Text Miner. We also created a term-by-frequency matrix by performing term identification. This step produced 604,085 terms from our review pool. Therefore, we needed another step to reduce the number of terms. In the term reduction step, we used the stop words list to reduce meaningless words (e.g., a, an, the, and etc.). We also handled the synonyms in the review texts. Finally, we used the Term Frequency-Inverse Document Frequency (TF-IDF) weighting option to place less weight on more frequent terms and more weight on less frequent terms. In the singular value decomposition step, we reduced the dimensionality of the term-by-frequency matrix by summing up the terms in groups. In the last step, the text cluster analysis, we performed a cluster analysis to classify each review into three categories: positive, negative, and other (e.g., irrelevant or neutral). We then used the distance-based clustering algorithm and created positive and negative sentiment groups. Then, based on the cosine similarity, each review was categorized into positive, negative, or other (e.g., irrelevant or neutral). That is, if one review doesn't have any positive or negative terms, then it belongs to the other category. To distinguish between the neutral reviews and the irrelevant ones from the other review text category, we manually classified other review texts. Our two research assistants manually checked all review texts and screened irrelevant review texts from other category. They worked independently in different locations, and thus, couldn't influence each other when they declared the irrelevant review texts. If the irrelevant categorization by the two research assistants was inconsistent, the inconsistent review texts were categorized by one of the authors of this study. We then separate irrelevant one from other category (either irrelevant or neutral ones) and then have neutral review text category.

The positive, negative, and neutral categories accounted for 42.64%, 12.65%, and 44.71%, respectively, of the total review texts. We then calculated the distribution of the user rating values that users ascribed to a movie while writing down review texts. Table 1

² To hold this argument, we note that the distribution of the user's numeric rating values must be different from that of the entropy of the review texts. As such, we conducted several tests and illustrated that the two sentiment distribution is different. For more detail, please see the data section.

³ For the period April 2014 through October 2014, we have collected all movies which were 204 movies.

and Fig. 2 present the distribution of the user rating values and the distribution of the review text sentiment (e.g., positive, negative, and neutral). We found that the proportion from 8 points up to 10 points, representing positive reviews, accounts for 71.36%, while the proportion of positive review sentiment from text-mining is around 42.64%. In addition, the proportion from 4 points up to 7 points, representing neutral reviews, accounts for 16.62%, while the proportion of neutral review sentiment from text-mining is around 44.71%. Finally, the proportion from 1 point up to 3 points, representing negative reviews, accounts for 12.02%, while the proportion of negative review sentiment from text-mining is around 12.65%. We observed that the proportion of negative review ratings is similar to the negative review text sentiment, but the distribution of positive and neutral ratings are totally different from that of the text-mining results. In addition, we conducted a two-sample *t*-test and variance ratio test to see whether the distribution of the entropy of the review text sentiment and numeric rating values. Both tests indicated that the distribution of the two variables were significantly different ($t = 111.284$ and $F = 45.810$, respectively). To see the moderating role of the entropy of the review text sentiment on the WOM effects, we used the results of our text classification analysis as an independent variable.

After the classification process was complete, we counted the numbers of each category (positive, negative, and neutral) on day *t* for individual movie *i*. To analyze the impact of the ratios of the review text sentiment on movie sales, we then calculated the entropy of the review text sentiment (positive, negative, and neutral) each day for each movie as follows: $\sum_i -p_i \log_2 p_i$ where: p_i is the proportion of class *i*. A relatively higher level of entropy means that the review text sentiment was distributed evenly, while a lower level of entropy indicates that the review text sentiment was biased towards a particular sentiment.

According to the previous studies, WOM is known to influence product sales in two ways: WOM volume and WOM valence (Liu, 2006). WOM volume indicates the amount of WOM exposed to consumers. WOM valence indicates the direction of consumer opinion about product quality, such as being positive or negative. Thus, we created WOM volume and valence as the total amount of the review texts and the mathematical average of the user rating values, respectively. When potential consumers search for online WOM information about movies, they may encounter the first page, which displays the total number of posted reviews and the average user rating values for movies. We first created $NUMOFREVIEW_{it}$ to verify the buzz effect on movie box office sales. When a consumer leaves a review comment⁴ about a movie on a Naver Movie, she or he has to give a rating value⁵ along with some review text. Thus, we created $AVGOFRATING_{it}$, which is the average value of the user ratings that occurred on the same day *t* for movie *i*. Table 2 provides the descriptions for our variables.

The control variable $SCREEN_{it}$ captures the exposure of a movie *i* on day *t*. The number of screens represents the exposure of a particular movie to prospective consumers. If they arrive at a theater and purchase a ticket for a movie, they may judge a movie's quality by how many screens are playing that movie. Of course, if they purchase a ticket in advance through the Internet or a mobile device, the more screens they see, the higher the perceived exposure. The dummy variable $WEEKEND_{it}$ was created to capture the weekend effect, as movie sales are expected to increase during the weekend. Table 3 presents the summary statistics for our variables. Finally, we created $SALES_{it}$ as our dependent variable representing movie *i*'s box office sales on day *t*. As prior studies have

Table 1
Distribution of user rating values.

Rating value	Category	Frequency	Percentage
10	Positive	120,054	46.71%
9	Positive	34,086	13.26%
8	Positive	29,256	11.38%
7	Neutral	16,902	6.58%
6	Neutral	11,463	4.46%
5	Neutral	9447	3.68%
4	Neutral	4891	1.90%
3	Negative	4489	1.75%
2	Negative	4153	1.62%
1	Negative	22,257	8.66%

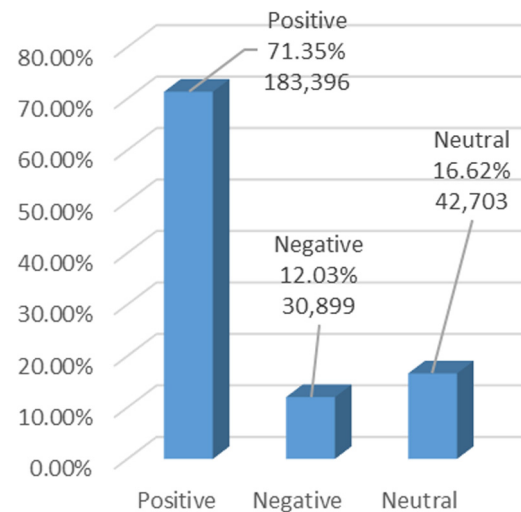


Fig. 2. Distribution of text sentiment categories.

Table 2
Variables.

Variable	Description
$SALES_{it}$	Movie sales(revenue) for movie <i>i</i> at day <i>t</i>
$ENTROPY_{it}$	Entropy of review text sentiments for movie <i>i</i> at day <i>t</i>
$AVGOFRATING_{it}$	Average rating values for movie <i>i</i> at day <i>t</i>
$NUMOFREVIEW_{it}$	Total number of reviews for movie <i>i</i> at day <i>t</i>
$SCREEN_{it}$	Daily number of screens for movie <i>i</i> at day <i>t</i>
$WEEKEND_{it}$	A dummy variable, =1 if day <i>t</i> is Friday, Saturday, Sunday =0 otherwise

shown a relationship between WOM and retail sales (Chevalier and Mayzlin, 2006), we used $SALES_{it}$ as the dependent variable for our model as well. In addition, we created the lag term ($SALES_{it-1}$) of our dependent variable to control for possible autocorrelation. We did not include the price of a movie as a control variable. The price usually influences the demand for a product, according to the market principle of supply and demand. Price may also play an important role in influencing consumer purchasing decisions. However, movie ticket prices are typically determined by the local markets in the same manner. Therefore, we can rule out the possibility that price is mediating the consumer purchasing decisions (Duan et al., 2008a).

We used a first-order lagged term for the WOM variables to exclude possible endogeneity between movie sales and WOM

⁴ The length of the reviews must be less than 140 Korean characters with spaces.

⁵ A rating value ranges from 1 to 10 points.

Table 3
Descriptive Statistics.

Variable	Obs.	Mean	Std. dev.	Min	Max
$SALES_{it}$	2856	231,000 [*]	7520,000 [*]	5 [*]	10,000,000 [*]
$ENTROPY_{it}$	2856	0.45	0.44	0	1
$AVGOFRATING_{it}$	2856	7.33	1.74	0	10
$NUMOFREVIEW_{it}$	2856	826.93	2,504.54	0	37,876
$SCREEN_{it}$	2856	156.39	251.09	1	1602
$WEEKEND_{it}$	2856	0.43	0.50	0	1
$RECENTRATING_{it}$	2856	7.52	2.98	1	10

^{*} Indicates 1000 Korean won (1000 won is around \$0.9 at our research period).

Table 4
Correlation.

Variable	[1]	[2]	[3]	[4]	[5]	[6]	[7]
[1] $SALES_{it}$	1.000						
[2] $ENTROPY_{it}$	0.295	1.000					
[3] $AVGOFRATING_{it}$	0.102	0.189	1.000				
[4] $NUMOFREVIEW_{it}$	0.697	0.298	0.055	1.000			
[5] $SCREEN_{it}$	0.810	0.522	0.135	0.693	1.000		
[6] $WEEKEND_{it}$	0.101	0.004	0.020	−0.011	0.021	1.000	
[7] $RECENTRATING_{it}$	−0.002	−0.022	0.447	−0.019	−0.031	−0.027	1.000

variables (Duan et al., 2008a). Table 4 provides the correlation statistics among the key variables.

5. Model Specification

The purpose of our study is to investigate the relationship between movie box office sales and the entropy of the review text sentiment. Thus, we developed the following equation. We applied a log-linear model in order to measure a percent change in our variables. To avoid possible multicollinearity among variables, all variables were mean centered.

$$\begin{aligned}
 SALES_{it} = & \alpha_1 ENTROPY_{i,t-1} + \alpha_2 ENTROPY_{i,t-1}^2 \\
 & + \alpha_3 AVGOFRATING_{i,t-1} + \alpha_4 NUMOFREVIEW_{i,t-1} \\
 & + \beta_1 (AVGOFRATING_{i,t-1} \times ENTROPY_{i,t-1}) \\
 & + \beta_2 (AVGOFRATING_{i,t-1} \times ENTROPY_{i,t-1}^2) \\
 & + \beta_3 (NUMOFREVIEW_{i,t-1} \times ENTROPY_{i,t-1}) \\
 & + \beta_4 (NUMOFREVIEW_{i,t-1} \times ENTROPY_{i,t-1}^2) \\
 & + \gamma_1 SALES_{i,t-1} + \gamma_2 SCREEN_{i,t-1} \\
 & + \gamma_3 WEEKEND_{it} + \gamma_4 RECENTRATING_{it} + \theta_t + \mu_i + \varepsilon_{it}
 \end{aligned}$$

where: $i = 1, \dots, N$ index of the movies. The dependent variable, $SALES_{it}$, denotes the standardized log value of box office sales of movie i on day t . Its first-order lagged variable is defined as $SALES_{i,t-1}$. $ENTROPY_{i,t-1}$ is the key variable that denotes the entropy level of the review text sentiment of movie i on day $t - 1$.

We expect that there is a non-linear relationship between the WOM variables and the entropy of the review text sentiment, so we used the squared terms between entropy and the WOM variables to examine their decreasing effects on movie box office sales. Thus, we created $ENTROPY_{i,t-1}^2$, a squared term of $ENTROPY_{i,t-1}$, to determine if the entropy level of the review text sentiment has a decreasing impact on movie box office sales when the entropy level becomes too high.

$AVGOFRATING_{i,t-1}$ indicates the arithmetic average of the user rating values for movie i on day $t - 1$, while $NUMOFREVIEW_{i,t-1}$ denotes the total number of reviews for movie i on day $t - 1$. The coefficients of $AVGOFRATING_{i,t-1}$ and $NUMOFREVIEW_{i,t-1}$ capture the valence and volume effects of the online WOM, respectively.

We then created four interaction terms to measure the interaction effects. $AVGOFRATING_{i,t-1} \times ENTROPY_{i,t-1}$ represents the interaction effect between the entropy level and the user rating values for movie i on day t . This interaction term suggests that the impact of the average user rating values on movie sales increases when the user ratings interact with a higher level of entropy. H1a suggests that the entropy level has a positive effect on movie sales when interacting with the WOM valence effect. Thus, we expect $\beta_1 > 0$. $AVGOFRATING_{i,t-1} \times ENTROPY_{i,t-1}^2$ captures the decreasing impact of the entropy on movie sales when interacting with the WOM valence effect. Based on H2a, we expect that β_2 will be negative. We also created $NUMOFREVIEW_{i,t-1} \times ENTROPY_{i,t-1}$ and $NUMOFREVIEW_{i,t-1} \times ENTROPY_{i,t-1}^2$ in the same manner. From H1b and H2b, we expect β_3 is to be positive and β_4 is to be negative.

$SCREEN_{i,t-1}$ denotes the number of cinema screens for movie i on day $t - 1$. The variable is included as a control variable in the equation. A dummy variable, $WEEKEND_{it}$, was included to control for the potential difference in consumer movie-going behavior between the weekend and the weekdays (Duan et al., 2008a). The variable, $RECENTRATING_{it}$, represents the first review page's rating values. As consumers usually couldn't read all the reviews, they were saliently exposed to the first page of reviews (Gu and Ye, 2014). We also include the fixed effects, μ_i , to control for the idiosyncratic properties related to each movie. For example, each movie has different characteristics (e.g., genre, recognition of the actors or directors of a movie, social or political issues in a movie). Likewise, we also included the time-fixed effects, θ_t , to capture the difference among the dates⁶ on which the movies were shown.

6. Results

6.1. Main results

We conducted a fixed-effect estimation for unbalanced panel data. Individual and time-fixed effects were included to capture the idiosyncratic attributes of each movie and each date, respec-

⁶ e.g., national holidays, election days, or whenever more consumers can go to a theater.

Table 5
Results.

Variable	Full equation	Without interactions	Without square terms
$ENTROPY_{i,t-1}$	-0.005 (0.003)	-0.006 (0.003)	-0.004 (0.003)
$ENTROPY^2_{i,t-1}$	-0.001 (0.002)	-0.003 (0.002)	
$AVGOFRATING_{i,t-1}$	0.001 (0.005)	0.003 (0.015)	0.003 (0.015)
$NUMOFREVIEW_{i,t-1}$	0.075*** (0.015)	0.078 (0.015)	0.075*** (0.015)
$AVGOFRATING_{i,t-1} \times ENTROPY_{i,t-1}$	0.011** (0.004)		0.010** (0.004)
$AVGOFRATING_{i,t-1} \times ENTROPY^2_{i,t-1}$	0.002 (0.002)		
$NUMOFREVIEW_{i,t-1} \times ENTROPY_{i,t-1}$	0.016*** (0.004)		0.015*** (0.003)
$NUMOFREVIEW_{i,t-1} \times ENTROPY^2_{i,t-1}$	0.000 (0.002)		
$SALES_{i,t-1}$	0.560*** (0.015)	0.560*** (0.015)	0.560*** (0.015)
$SCREEN_{i,t-1}$	0.037*** (0.005)	0.036*** (0.005)	0.038*** (0.005)
$WEEKEND_{it}$	4.975*** (0.437)	4.799*** (0.433)	4.994*** (0.436)
$RECENTRATING_{it}$	0.000 (0.001)	0.000 (0.001)	0.000 (0.001)
Group fixed effect	Yes	Yes	Yes
Time fixed effect	Yes	Yes	Yes
R-square	Within = 0.9754 Between = 0.0527 Overall = 0.2248	Within = 0.9751 Between = 0.0549 Overall = 0.2371	Within = 0.9754 Between = 0.0529 Overall = 0.2239
Number of observations	2379	2379	2379

*p < 0.5, **p < 0.01, ***p < 0.00.

Note: Individual dummies and time dummies used in estimation but not reported.

tively. Column 1 in Table 5 provides the estimation results of the full equation. The results show that the entropy of the review text sentiment does not influence movie sales by itself (neither does its squared term). This supports our arguments that consumers usually have different opinions about the same movie, since a motion picture is a preference good. Thus, it is natural that there are various sentiments (positive, negative, and neutral) in review texts. Therefore, consumers may not be influenced by entropy itself. That is, when consumers decide which movie they want to watch, they usually simultaneously consider the movie rating values, number of the reviews, and review texts.

However, as illustrated in Table 5, the coefficient of $AVGOFRATING_{i,t-1} \times ENTROPY_{i,t-1}$ was positive and statistically significant. This result suggests that the entropy level of the review text sentiment was positively moderated with the WOM valence to influence movie sales. Thus, this result supports H1a.

Potential consumers are often exposed to a variety of review text sentiment, as movies are preference goods. If there are too many positive reviews, consumers may feel that the reviews or opinions are manipulated and less reliable. Therefore, when a movie is rated higher, consumers may have enhanced trust toward the movie's quality and be more likely to watch that movie, if they observe various types of review sentiment. Thus, the entropy can only boost movie box office sales when interacting with the WOM valence.

In addition, the coefficient of $NUMOFREVIEW_{i,t-1} \times ENTROPY_{i,t-1}$ was positive and statistically significant. This result suggests that the entropy of the review text sentiment was positively related to the WOM volume to influence movie sales. This result supports H1b. When a movie has a large number of reviews, consumers are more likely to trust the information and watch that movie, as the entropy of the review sentiment increases. Consumers may also more easily find proper reviews if the WOM volume is much larger.

It is natural that there are more of a variety of types of review text sentiment as the WOM volume increases, unless some related parties manipulate the reviews.

We also found that neither the coefficients of $AVGOFRATING_{i,t-1} \times ENTROPY^2_{i,t-1}$ nor $NUMOFREVIEW_{i,t-1} \times ENTROPY^2_{i,t-1}$ have significant coefficients in our empirical model. Thus, H2a and H2b were not supported. This implies that there are no decreasing effects of the entropy of the review text sentiment with both the WOM valence and the WOM volume when influencing movie box office sales. This suggests that consumers may not become confused, even though they observe too many negative or neutral review texts, given the same amount of positive reviews. That is, consumers may assign greater credibility to online WOM information (e.g., rating values or number of the reviews) until the ratio of positive, negative and other types of review text sentiment becomes equal.

The coefficient of $AVGOFRATING_{i,t-1}$, however, was not statistically significant, while $NUMOFREVIEW_{i,t-1}$ was statistically significant. These findings are consistent with that of previous studies. We also found that the other control variables have positive and statistically significant coefficients, as we expected.

6.2. Robustness check

To assess the robustness of our research model results, we estimated two more equations to examine possible multicollinearity. First, we estimated a model without any interaction terms, as they often cause multicollinearity problems. Second, we estimated another model without a squared term of entropy and its interaction terms. As we see in Columns 2 and 3 of Table 5, excluding all interaction terms does not influence the estimation results significantly. Only small changes in the figures and no changes in the

Table 6
Robustness check with entropy using numeric user rating values.

Variable	Full Equation	Without interactions	Without square terms
$NUMERICENTROPY_{i,t-1}$	0.073 (0.056)	0.079 (0.056)	.018 (.015)
$ENTROPY^2_{i,t-1}$	–0.053 (0.054)	–0.056 (0.053)	
$AVGOFRATING_{i,t-1}$	–0.007 (0.012)	0.003 (0.004)	.003 (.011)
$NUMOFREVIEW_{i,t-1}$	0.128*** (0.018)	0.076*** (0.014)	.124*** (.017)
$AVGOFRATING_{i,t-1} \times NUMERICENTROPY_{i,t-1}$	0.081 (0.049)		–.001 (.013)
$AVGOFRATING_{i,t-1} \times NUMERICENTROPY^2_{i,t-1}$	–0.078 (0.045)		
$NUMOFREVIEW_{i,t-1} \times NUMERICENTROPY_{i,t-1}$	0.068 (0.045)		.065** (.012)
$NUMOFREVIEW_{i,t-1} \times NUMERICENTROPY^2_{i,t-1}$	0.004 (0.042)		
$SALES_{i,t-1}$	0.557*** (0.014)	0.561*** (0.014)	.561*** (.014)
$SCREEN_{i,t-1}$	0.035*** (0.005)	0.035*** (0.005)	.033*** (.005)
$WEEKEND_{it}$	4.929*** (0.430)	4.974*** (0.434)	4.915*** (.430)
$RecentRating_{it}$	0.001 (0.001)	0.001 (0.001)	.001 (.001)
Group fixed effect	Yes	Yes	
Time fixed effect	Yes	Yes	
R-square	Within = 0.9747 Between = 0.044 Overall = 0.2294	Within = 0.9741 Between = 0.0413 Overall = 0.2255	Within = 0.9746 Between = 0.0434 Overall = 0.2297
Number of observations	2379	2379	2379

*p<0.5, **p<0.01, ***p<0.00.

Note: Individual dummies and time dummies used in estimation but not reported.

signs (positive and negative) of the estimation results existed when we added or removed some of the variables. Therefore, there is little multicollinearity concern in our research model.

As another robustness check, we created the entropy value using only the numeric ratings and re-estimated our research model. If the numeric user rating value had the same sentiment distribution with the text sentiment distribution, the entropy of the review sentiment did not moderate with the WOM information to influence movie sales. Thus, if a user rating value is from 1 to 5, we assigned it to a negative category; if a user rating value is from 6 to 8, we assigned it to a neither positive nor negative category; and if a user rating value is from 9 to 10, we assigned it to a positive category. By doing so, we could calculate the entropy value using only the numeric ratings.

Then we replaced our entropy variable using only review texts with the one using numeric ratings only and re-estimated our research model⁷. We found that the entropy value using the numeric ratings and its interaction terms was not statistically significant (see Table 6 for the detail).

This finding suggests that the entropy of the review text sentiment moderates the WOM effects, while the entropy of the numeric ratings does not. This supports our argument that movie consumers may have stereotypes towards WOM information, in particular, when they observe too many positive reviews. However, such a stereotype can be weakened when consumers observe an equally distributed review text sentiment. That is, when WOM valence and volume interact with a high level of entropy in the review text sentiment, the influence of the WOM information on product sales becomes stronger. Therefore, we show that there is a moderating effect of the entropy of the review text sentiment

on the relationship between the WOM effects (e.g., numeric rating values or number of the reviews) and product sales.

We also estimate a random effect model to examine which model was suitable for our dataset. The estimation results of the random effect model illustrated fewer differences from those of the fixed effect model. The results are shown in Table 7. The random effect model also supported our hypotheses. The key variables that have significant coefficients in the random effect model are similar to those of the fixed effect model. We found few changes in the coefficients in the figures between the fixed and random effect models, but these were not very remarkable with respect to the P-values.

We then conducted Hausman Specification Tests (Hausman, 1978), where the null hypothesis is that the preferred model is a random effect, versus the alternative, which is a fixed effect. The results of the tests indicated that the preferred model is the fixed effect model in the three cases by rejecting the null hypotheses (p-values for the Full equation, Without Interaction Terms, and Without Squared Terms, are all less than 0.000). This allows us to use the fixed effect model for our research, rather than the random effect model.

7. Conclusion

The primary objective of this study was to investigate the relationship between the entropy of the review text sentiment and the online WOM effects. Thus, we empirically illustrated that the entropy of the review text sentiment was positively moderated with WOM information (e.g., valence or volume) to influence movie box office sales.

We conducted text-mining techniques to classify review texts into three types of sentiment (positive, negative, and neutral). Subsequently, we calculated entropy levels with the classification

⁷ We also used the standard deviation of rating values as a moderator. However, we still found that its interaction terms are not statistically significant. We thank the anonymous reviewer for suggesting this robustness check.

Table 7
Robustness Check Between Random and Fixed Effect Models.

Variable	Full Equation		Without interactions		Without square terms	
	Fixed	Random	Fixed	Random	Fixed	Random
$ENTROPY_{it-1}$	-0.005	-0.008*	-0.006	-0.008*	-0.004	-0.008*
$ENTROPY^2_{it-1}$	-0.001	-0.002	-0.003	-0.001		
$AVGOFRATING_{it-1}$	0.001	-0.006	0.003	-0.002	0.003	-0.003
$NUMOFREVIEW_{it-1}$	0.075***	0.091***	0.078**	0.094***	0.075***	0.092***
$AVGOFRATING_{it-1} \times ENTROPY_{it-1}$	0.011**	0.011**			0.010**	0.010**
$AVGOFRATING^2_{it-1} \times ENTROPY^2_{it-1}$	0.002	0.003				
$NUMOFREVIEW_{it-1} \times ENTROPY_{it-1}$	0.016***	0.008*			0.015***	0.008**
$NUMOFREVIEW_{it-1} \times ENTROPY^2_{it-1}$	0.000	0.000				
$SALES_{it-1}$	0.560***	0.612***	0.560***	0.609***	0.560***	0.612***
$SCREEN_{it-1}$	0.037**	0.007	0.036**	0.007	0.038**	0.008
$WEEKEND_{it}$	4.975***	0.292	4.799**	0.288	4.994***	0.289
$RECENTRATING_{it}$	0.000	0.000	0.000	0.000	0.000	0.000
R-square	0.9754	0.9728	0.9751	0.9726	0.9754	0.9728
Within	0.0527	0.9752	0.0549	0.9752	0.0529	0.9751
Between	0.2248	0.9721	0.2371	0.9719	0.2239	0.9721
Overall	2379	2379	2379	2379	2379	2379
Number of observations						

* $p < 0.5$, ** $p < 0.01$, *** $p < 0.00$.

Note: Individual dummies and time dummies used in estimation but not reported.

results and included them in our dataset as a key variable. Using a fixed-effects estimation, we found that the entropy of the review text sentiment positively moderates the WOM variables when influencing the movie box office sales. These results suggest that consumers are more likely to trust WOM information when they equally encounter positive, negative, and neutral review texts. This comes about because consumers usually have and expect others to have different preferences about the same movie, and thus, see various types of the review text sentiment about a movie as credible and natural.

The findings of this study have several important implications for the extant literature. First, we show that the entropy of the review text sentiment has an impact on WOM effects, which boosts movie box office sales. This indicates that if a highly rated movie illustrates a variety of sentiment in the review texts, consumers are more likely to have credibility in the rating values or the number of the reviews, and thus, choose to watch the movie due to the dilution effect. Additionally, if a movie with many more reviews displays a variety of sentiment in the review texts, consumers may believe that the reviews are written by real customers, rather than by hired writers from related parties. In conclusion, the entropy of the review text sentiment has a positive impact on WOM effects in boosting movie box office sales. Second, prior studies showed that user ratings do not have a significant impact on product sales (Duan et al., 2008a; Gu et al., 2012). That is, user rating values may not play a dominant role as an indicator of WOM valence. Likewise, our results are similar to prior studies that illustrated an insignificant relationship between user ratings and movie sales. We suggest that these controversial results may come from the lack of consideration of the review text sentiment and their moderating relationship with WOM information. Thus, we empirically showed that user ratings have a significant impact on movie sales when interacting with the entropy of the review text sentiment. As such, we believe that our results suggest one of the possible answers for this controversial issue.

The results have practical implications. Too many positive reviews may limit consumer purchasing decisions. People have different preferences about movies, so the existence of positive, negative or other types of review texts among review pools seems natural to consumers. If consumers encounter too many positive reviews about a movie, they may think that the reviews are being manipulated, and thus, may be reluctant to choose the movie. Therefore, online retailers or related parties must reconsider the practice of hiring someone to post positive reviews about a movie or to delete negative reviews, as our results illustrate that censoring negative ratings just decreases the entropy value, which leads to a less influential WOM effect. Consequently, from a managerial perspective, deleting negative reviews or including more positive reviews may not help online retailers or related parties.

This study has some limitations. First, we did not directly observe how potential consumers value or perceive the WOM information, given the various types of review text sentiment ratios. It would be helpful if future researchers can conduct field experiments to investigate how the user perception about the value of the WOM increases (or decreases), depending on the ratio of the review text sentiment. Second, our analysis is limited to online users who left reviews at a Korean movie review website (e.g., Naver movie). Hence, our analysis focuses on movies released in South Korea and review texts written in Korean. It would be interesting if future research to expand our study to a worldwide context or at least to other languages. Third, our research was limited to the movie industry. We found that positive reviews are dominant in the movie industry. However, other industries may intervene when there are multiple interest groups. Due to the Negativity Bias, in which people are more willing to express negative opinions than positive ones (Rozin and Royzman, 2001), neg-

ative reviews are usually easy to find in other industries. Future research that investigates the impact of the ratios in the review text sentiment in other industries would be helpful. Fourth, online users may not read all reviews. Thus, the entropy value of all reviews may not be used by them. Although we added first page of the review ratings as a control variable to see how it influences the results, we think that the entropy values may vary across review pages. Finally, we leave other researchers to analyze the impact of irrelevance in review texts due to the limitation of text-mining techniques. Additionally, analyzing the impact of abstractness in reviews seems to be an interesting issue. Many consumers leave abstract comments. Therefore, researchers may ask whether potential consumers are more attracted to concrete or sincere reviews. Future research can examine the impact of abstractness and the concreteness of the reviews and movie sales.

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